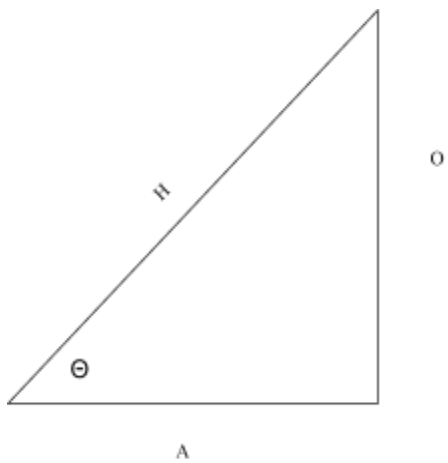
Analytic Trig

Up until now, our understanding of trigonometry has primarily been focused on triangles, with a bit of the Unit Circle sprinked in. Of course, there's a lot more to trigonometry than that.

Trigonometric identities are equations that relate trigonometric functions, and hold true for any value Θ, assuming both sides of the equality are defined -- for example, you can't have $sec\left(\frac{\pi}{2}\right)$, or $tan\left(\frac{\pi}{2}\right)$, or $csc(-\pi)$. From a geometric standpoint, these identities, if equal, hold true for any right triangle.

Where do they come from? Well, some of them can actually be derived directly from a right triangle. For example:



Looking at the diagram to the left, we know that $sin\,(\Theta) = \frac{O}{H}$ and $cos\,(\Theta) = \frac{A}{H}$.

Accordingly, it follows that

$$sin^2\,(\Theta) + cos^2(\Theta) = \frac{O^2}{H^2} + \frac{A^2}{H^2}$$

$$sin^2\,(\Theta) + cos^2(\Theta) = \frac{O^2 + A^2}{H^2}$$

Since we know by the Pythagorean Theorem that for any right triangle, $O^2 + A^2 = H^2$, we can say that:

$$sin^2\,(\Theta) + cos^2(\Theta) = \frac{H^2}{H^2} = 1.$$

This identity -- $sin^2\,(\Theta) + cos^2(\Theta) = 1$ -- is one of the most widely seen trigonometric identities (it is actually 1 of 3 *Pythagorean Identities*, which we will of course cover) , and can be used to derive many other identities.

Lots of these equations are formulated from the unit circle and triangles. If you are interested in why the following equations work, I highly recommend looking it up on your own and studying them.

Before you do anything else, it would be useful to memorize the following definitions below:

$$tan\,(\Theta) = \frac{sin\,(\Theta)}{cos\,(\Theta)} \quad \textbf{(1)}$$

$$cot\,(\Theta) = \frac{cos\,(\Theta)}{sin\,(\Theta)} \quad \textbf{(2)}$$

$$csc\,(\Theta) = \frac{1}{sin\,(\Theta)} \quad \textbf{(3)}$$

$$sec\,(\Theta) = \frac{1}{cos\,(\Theta)} \quad \textbf{(4)}$$

<u>Pythagorean Identities:</u>

Pythagorean Identities are just simply means of expressing the Pythagorean Theorem as trigonometric functions. You already know the first one: $sin^2(\Theta) + cos^2(\Theta) = 1$. From here, we will derive the other 2 identities.

1. $sin^2(\Theta) + cos^2(\Theta) = 1$
2. $tan^2(\Theta) + 1 = sec^2(\Theta)$

We can easily see that this is true using the first identity to verify this identity:

$tan^2(\Theta) + 1 = sec^2(\Theta)$

$\frac{sin^2(\Theta)}{cos^2(\Theta)} + \frac{cos^2(\Theta)}{cos^2(\Theta)} = sec^2(\Theta)$

$\frac{sin^2(\Theta) + cos^2(\Theta)}{cos^2(\Theta)} = sec^2(\Theta)$

$\frac{1}{cos^2(\Theta)} = sec^2(\Theta)$

$sec^2(\Theta) = sec^2(\Theta)$

3. $cot^2(\Theta) + 1 = csc^2(\Theta)$

Again, we can easily see that this is true using the first identity to verify this identity:

$cot^2(\Theta) + 1 = csc^2(\Theta)$

$\frac{cos^2(\Theta)}{sin^2(\Theta)} + \frac{sin^2(\Theta)}{sin^2(\Theta)} = csc^2(\Theta)$

$\frac{cos^2(\Theta) + sin^2(\Theta)}{sin^2(\Theta)} = csc^2(\Theta)$

$\frac{1}{sin^2(\Theta)} = csc^2(\Theta)$

$csc^2(\Theta) = csc^2(\Theta)$

So, besides the definitions, all you need to memorize are

$sin^2(\Theta) + cos^2(\Theta) = 1$ **(5)**

$tan^2(\Theta) + 1 = sec^2(\Theta)$ **(6)**

$cot^2(\Theta) + 1 = csc^2(\Theta)$ **(7)**

By converting the identities and definitions above, and manipulating them accordingly, you can simplify or verify almost any simplifying trig question by using the chart above.

1. Let's start with an easy one: $tan(\Theta)cos(\Theta)$

How could we simplify this? Well, remember that $tan\ (\Theta) = \frac{sin\ (\Theta)}{cos\ (\Theta)}$, so really, our expression is just:

$\frac{sin\ (\Theta)}{cos\ (\Theta)}\ cos(\Theta)$

Which is just equal to $sin(\Theta)$. Easy enough, right?

2. This one is a little bit harder: $cos(\Theta) + tan(\Theta)\ sin(\Theta)$

Using the 7 identities we already know, we can rewrite $cos(\Theta) + tan(\Theta)\ sin(\Theta)$ as

$cos(\Theta) + \frac{sin^2\ (\Theta)}{cos\ (\Theta)} = \frac{cos^2\ (\Theta)}{cos\ (\Theta)} + \frac{sin^2\ (\Theta)}{cos\ (\Theta)} = \frac{sin^2\ (\Theta) + cos^2(\Theta)}{cos\ (\Theta)} = \frac{1}{cos\ (\Theta)} = sec(\Theta)$.

So, $cos(\Theta) + tan(\Theta)\ sin(\Theta)$ is just equal to $sec(\Theta)$.

Try to simplify the expressions below:

1. $sec^2(\Theta)$ - 1
2. $\sin(\Theta)\cot(\Theta)$
3. $\frac{sin^2\ (\Theta)}{1+cos(\Theta)}$
4. $\frac{cot^2(\Theta)}{1+csc(\Theta)}$
5. $csc^2(\Theta)$ - $cot^2(\Theta)$ + 1
6. $\frac{sin(\Theta) + cos(x)}{sin(\Theta)cos(\Theta)}$

We can also *verify* trigonometric identities by simplifying one side of a trigonometric equation to show that both sides are equal:

For example, let's say we have this equation.

$sec(x)cot(x) = csc(x)$

By plugging in the top identities into this equation we can simplify it.

Since, $sec(x) = \frac{1}{cos(x)}$, $cot(x) = \frac{cos(x)}{sin(x)}$, $csc(x) = \frac{1}{sin(x)}$

Therefore, $\frac{1}{cos(x)} * \frac{cos(x)}{sin(x)} = \frac{1}{sin(x)}$. The cos(x) cancels which leaves you with, $\frac{1}{sin(x)} = \frac{1}{sin(x)}$ or $\frac{1}{sin(x)} = csc(x)$.

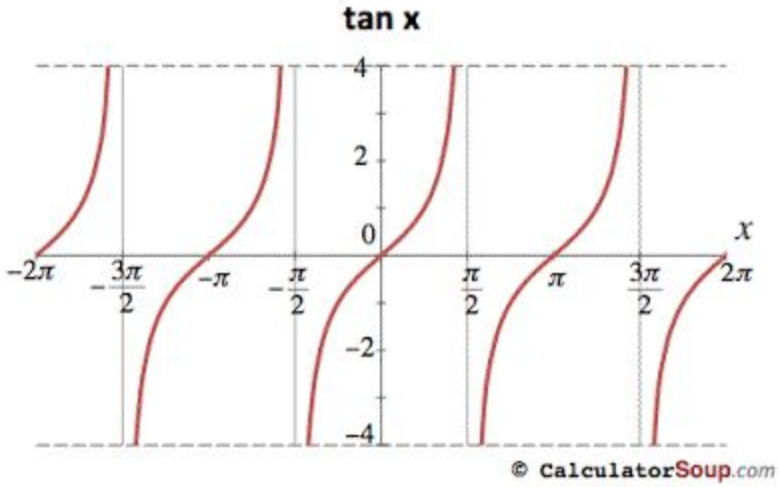
So, we finish with $csc(x) = csc(x)$, which of course holds.

Graphing *tan (x), cot (x), sec (x), csc (x)* Functions.

These graphs are not like sin and cos because these graphs have discontinuities.

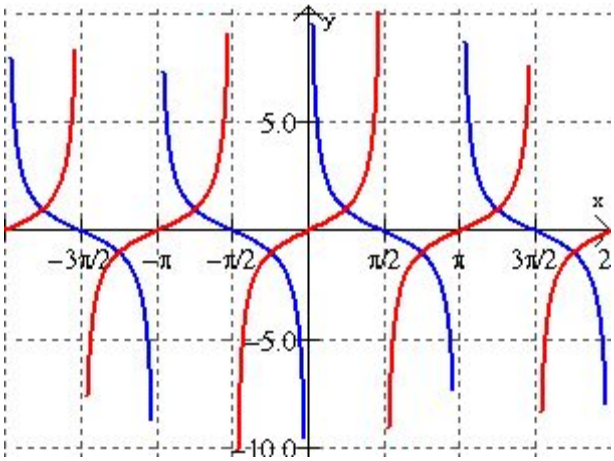
1. *tan* (x)

The equation for the tan function is $y = a * tan(bx + c) + d$ where a is the amplitude, b is the frequency, c is the phase shift, and d is the vertical displacement. Also, you must note that unlike sin and cosine, the period of tan is not 2pi but rather just pi.



2. *cot* (x)

This is the most similar to the tan function because it is tan but inversed. Remember that $cot(x) = \frac{1}{tan(x)}$. The equation for the tan function is $y = a * cot(bx + c) + d$ where a is the amplitude, b is the frequency, c is the phase shift, and d is the vertical displacement. The shape of the cot function looks flipped compared to the tan function and is misplaced a little. As a result, the key points for both equations are different as shown below.



The red is the tan function while the blue is the cot function.

Key Points

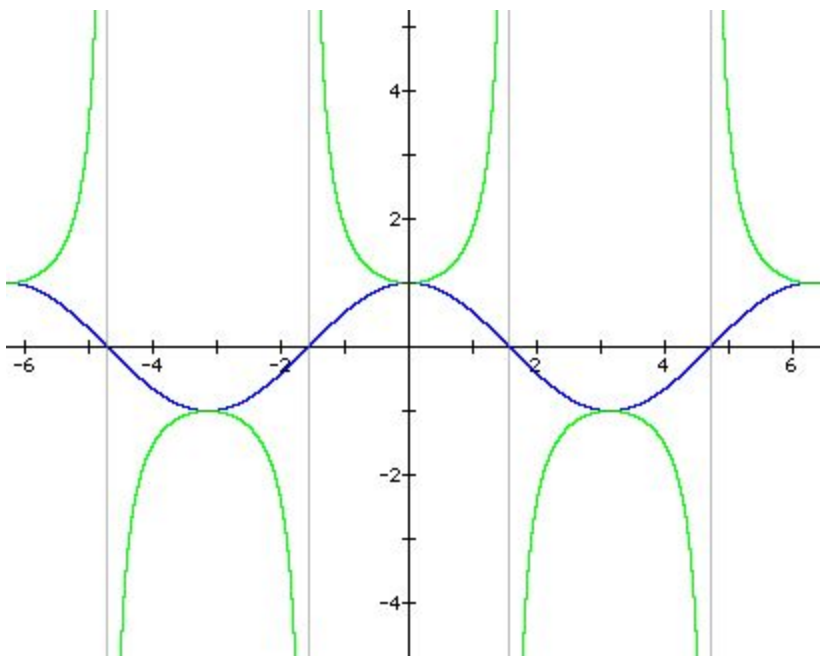
x	tan x
0	0
π/4	1
π/2	UD
3π/4	-1
π	0

x	cot x
0	UD
π/4	1
π/2	0
3π/4	-1
π	UD

Memorize the key points for both tan and cot functions.

3. *sec* (x)

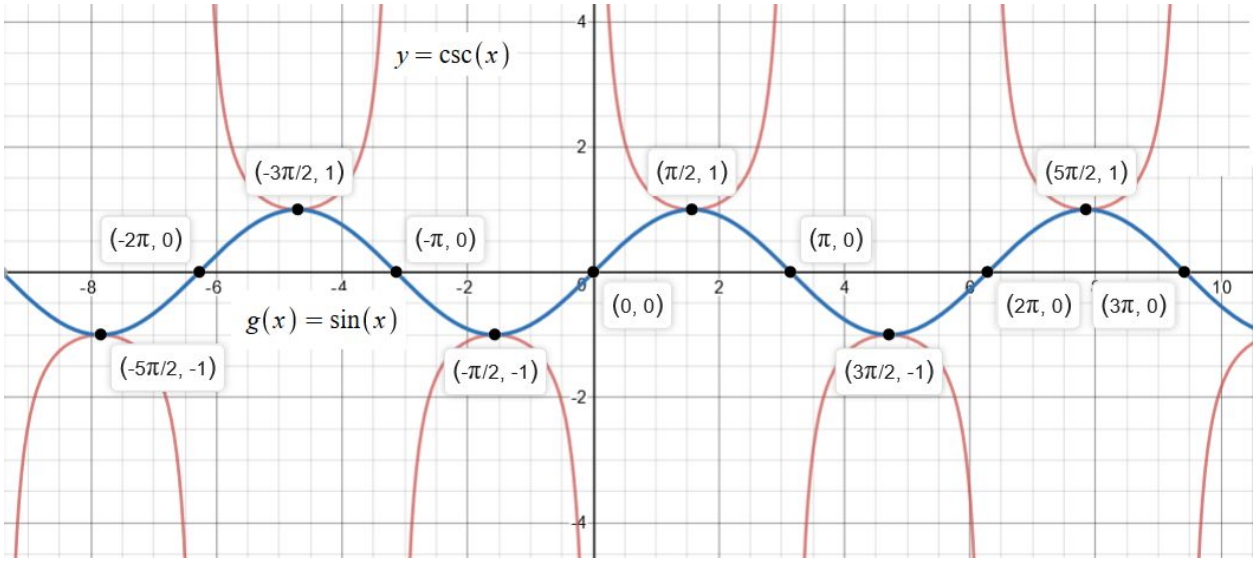
Since sec is the inverse of the cosine function, finding the secant function becomes a lot easier.



The blue is cosine while the green is the secant function. The secant function also has discontinuities. This secant has discontinuities every pi starting at pi/2.

4. *csc* (x)

This is similar to the shape of secant because it is the inverse of the sin function.



The blue is the sin function while the red is the csc function. For this csc function every pi there is a discontinuity.

I highly recommend using desmos to mess around with these functions.

Applications of Trigonometry:

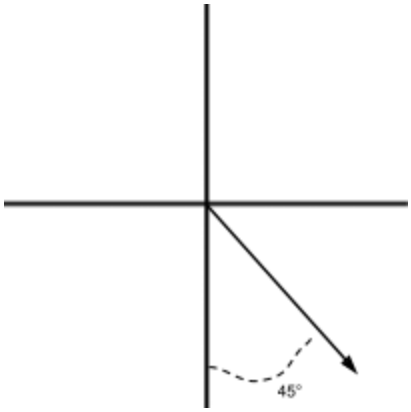
In trigonometry, right triangles may be used to find distances through angles that are shown as 'bearings'.

- A **bearing** is thought to be the direction from an object to another (typically in navigation). These bearings are given as angles that are rotated clockwise from the north side.
- Strict definition of **bearing:** bearing is how direction is measured at sea. North is 0*, east is 90*, south is 180*, and west is 270*.

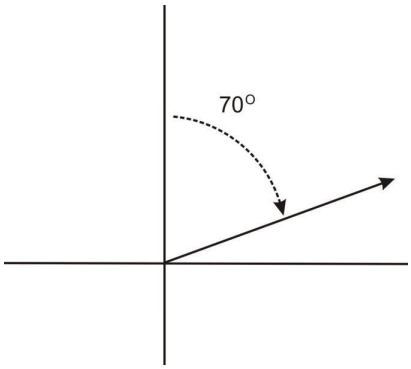
<u>Bearings start in the direction of the first one and fall in the direction of the second</u>

For example, the bearing **S45°E** means that the direction is 34° away from due south directed towards the east.

<u>You start graphing from the south direction and graph 45° degrees towards the east.</u>



The picture on the right is an example of **N70°E**. It is the angle 70 degrees from north (positive y-axis) to east (positive x-axis). <u>You start graphing from the north and graph 70 degrees towards the east.</u>

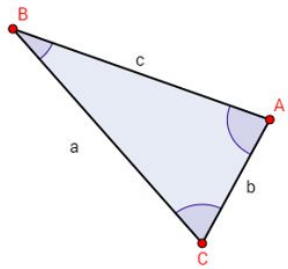


If you were to see **N70°W**, it would mean 70 degrees from the north to west (towards negative x-axis). <u>You start from the north and graph 70 degrees towards the west.</u>

Let's do a word problem using bearings:

This word problem requires knowledge of triangle formulas, specifically the law of cosines:

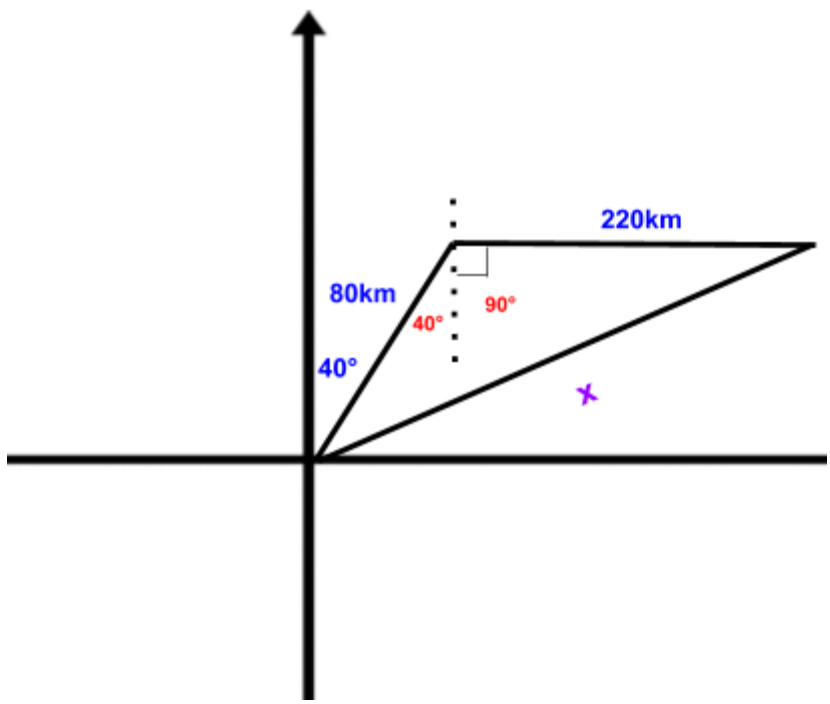
Law of Cosines



$$a^2 = b^2 + c^2 - 2bcCosA$$
$$b^2 = a^2 + c^2 - 2acCosB$$
$$c^2 = a^2 + b^2 - 2abCosC$$

A plane flies at a bearing of N40°E for 80km, it then changes bearings and flies directly east for 220km. How far is the plane from where it started?

We begin by drawing a picture representing the word problem:



What's written in **blue** is **given** in the word problem

What's written in **red** is what we can **figure out** using the problem. The 40 degree angle given in the problem and the 40 degree angle we figured out are vertical angles, so they have the same degree. When the plane flew directly west, a 90 degree angle was created.

To figure out **how far the plane is from where it started**, we need to find the **"x"** labeled in the picture. This involves the use of the law of cosines:

$$x^2 = 80^2 + 220^2 - (2 * 80 * 220cos(130))$$

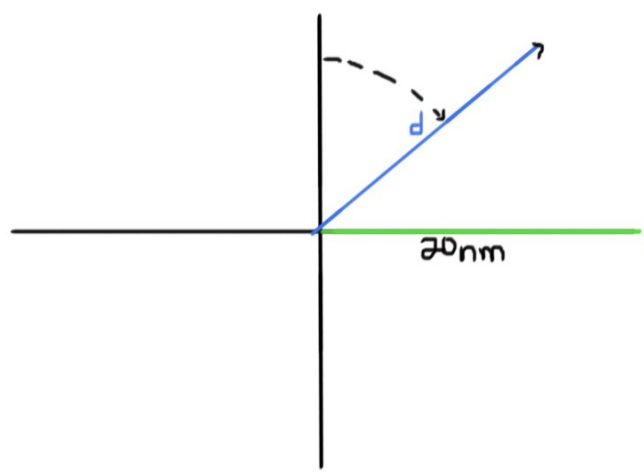
$$x = \sqrt{80^2 + 220^2 - (2 * 80 * 220cos(130))}$$

Plugging this into the calculator, we get:

x =278.255 km

Using this knowledge, let's look at an example problem:

A ship travels on a N60*E course. The ship travels until it is due north of a port which is 20 nautical miles due east of the port from which the ship started. How far did the ship travel?



The angle in between d and 20nm is known as the complement of 60*, this would be 30*. So, we can find d using the cosine function:

$$cos(30\ ) = \frac{adjacent}{hypotenuse} = \frac{20}{d}$$

$$cos(30\ ) = \frac{20}{d}$$

$$d\ cos(30) = 20$$

$$d = \frac{20}{cos(30)} = 129.65$$